Key Ideas Reference

🔑 Key Ideas — End-of-Year Maths Reference Sheet

Student: Shrishti — Year 8, St Paul's Girls School
Use: Quick reference of every rule, formula and definition for revision.
Tip: Don't try to memorise it all at once — read it through, then use it as a look-up sheet while you practise.

🧮 ALGEBRA

Inequalities

- Solve like an equation — do the same operation to both sides.
- **Flip the sign** when you multiply or divide by a **negative** number (`<` ↔ `>`, `≤` ↔ `≥`).
- Number-line notation: ○ = endpoint **not** included (`<`, `>`); ● = endpoint **included** (`≤`, `≥`).
- Shading: left for "less than", right for "greater than".
- For integer solutions, solve first, then list only whole numbers that fit.

Indices (powers)

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^0 = 1$ (anything to the power 0 is 1)
- $a^{-n} = 1 / a^n$ (negative power = reciprocal)
- $a^{(1/n)} = \sqrt[n]{a}$, $a^{(m/n)} = (\sqrt[n]{a})^m$
- $(ab)^n = a^n b^n$, $(a/b)^n = a^n/b^n$

Expansion of brackets

- **Single bracket:** $a(b + c) = ab + ac$
- **Double bracket (FOIL):** $(x + a)(x + b) = x^2 + (a+b)x + ab$
- **Squaring a bracket:** $(x + a)^2 = x^2 + 2ax + a^2$
- **Difference of two squares:** $(x + a)(x - a) = x^2 - a^2$

Factorisation

- **Common factor first:** $6x^2 + 9x = 3x(2x + 3)$.
- **Quadratic $x^2 + bx + c$:** find two numbers that **multiply to c** and **add to b**.
- **Difference of two squares:** $x^2 - a^2 = (x + a)(x - a)$.
- **Quadratic with leading coefficient a ($ax^2 + bx + c$):** split the middle term — find two numbers that multiply to $a \times c$ and add to b.

Quadratic equations

- Standard form: $ax^2 + bx + c = 0$.
- **Solve by factorising:** factor, then set each bracket = 0.
- **Quadratic formula:**

  $x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$
- The **discriminant** $b^2 - 4ac$ tells you how many real solutions:
  - > 0 → two solutions
  - $= 0$ → one repeated solution

- `< 0` → no real solutions

Surds

- `√(ab) = √a × √b`, `√(a/b) = √a / √b`
- Simplify by pulling out the largest **square** factor: `√50 = √(25 × 2) = 5√2`.
- Add/subtract only **like surds**: `3√2 + 5√2 = 8√2`.
- **Rationalise the denominator:** multiply top and bottom by the surd, e.g. `1/√3 = √3/3`.

Sequences & nth term

- **Linear (arithmetic):** common difference `d`. nth term = `a + (n − 1)d` (or `dn + c`).
- **Quadratic:** second differences are constant. nth term contains an `n²` term; the coefficient of `n²` = (second difference) ÷ 2.
- **Geometric:** common ratio `r`. nth term = `a × r^(n − 1)`.
- To check a value is in the sequence, set the nth term = that value and solve for `n`; `n` must be a positive whole number.

Line equations (straight lines)

- **Slope-intercept form:** `y = mx + c`, where `m` = gradient, `c` = y-intercept.
- **Gradient between two points:** $m = (y_2 - y_1) / (x_2 - x_1)$.
- **Parallel lines** have the **same gradient**.
- **Perpendicular lines:** gradients multiply to `−1` (one is the **negative reciprocal** of the other).
- **Intersection of two lines:** solve the equations simultaneously.
- A line through (x_1, y_1) with gradient `m`: $y - y_1 = m(x - x_1)$.

Trial and Improvement

- Find two integers where the function is **too low** for one and **too high** for the other — the root is between them.
- Try values to 1 d.p. between them; track each trial in a **table**.
- Decide each result is "too high" or "too low" and narrow the interval.
- **To round to 1 d.p.:** test the midpoint (e.g. 3.55); if too low, the answer rounds **up** to 3.6.

📐 GEOMETRY

Congruence and Similarity

- **Congruent** = same shape **and** size (allowed: rotate / reflect / translate).
- **Similar** = same shape, possibly different size; corresponding angles equal, corresponding sides in the same ratio.
- **Triangle congruence rules:** SSS, SAS, ASA / AAS, RHS.
- **Scale factors for similar shapes:**
  - Length scale factor: `k`
  - Area scale factor: `k²`
  - Volume scale factor: `k³`
- Always match **corresponding** sides before writing a ratio.

Loci & Constructions

- **Distance r from a point** → circle of radius r.
- **Distance r from a line segment** → "stadium": parallel lines + semicircles at ends.
- **Equidistant from two points** → perpendicular bisector of the segment joining them.
- **Equidistant from two lines** → angle bisector.
- **Constructions** (compass + ruler only):

- Perpendicular bisector: equal arcs from each endpoint, join intersections.
- Angle bisector: arc cuts both arms; equal arcs from those points; join.
- Perpendicular from a point to a line: arcs from the point cut the line twice; bisect those.
- Equilateral triangle: arcs of radius AB from A and B; join intersection.

- For **region** problems: draw all boundaries, decide which side satisfies each condition, shade the overlap.

Circle Theorems

1. **Angle at the centre = 2 × angle at the circumference** (same arc/chord).
2. **Angles in the same segment are equal.**
3. **Angle in a semicircle = 90°** (Thales).
4. **Opposite angles of a cyclic quadrilateral add to 180°.**
5. **Tangent ⊥ radius** at the point of contact.
6. **Two tangents from an external point are equal in length.**
7. *(Stretch)* **Alternate segment theorem:** angle between tangent and chord = angle in the opposite segment.

Circles — measurements

- Circumference: `C = 2πr = πd`
- Area: `A = πr²`
- **Arc length** = `(θ / 360) × 2πr`
- **Sector area** = `(θ / 360) × πr²`
- **Sector perimeter** = arc length + 2r
- **Semicircle:** half the full area / circumference, plus the diameter for perimeter.
- **Annulus (ring):** big circle area − small circle area.

Geometry using algebra

- Set up an equation using a known fact, then solve:
  - **Angles on a straight line** = 180°.
  - **Angles around a point** = 360°.
  - **Angles in a triangle** = 180°; **in a quadrilateral** = 360°.
  - **Sum of interior angles of an n-gon** = `(n − 2) × 180°`.
  - **Exterior angle of a regular n-gon** = `360° / n`.
- **Pythagoras:** `a² + b² = c²` (c = hypotenuse).
- **Isosceles** triangles → two equal sides ⇒ two equal base angles.

⚖️ NUMBER, RATIO & PROPORTION

Converting Units

- **Length:** 10 mm = 1 cm, 100 cm = 1 m, 1000 m = 1 km.
- **Mass:** 1000 mg = 1 g, 1000 g = 1 kg, 1000 kg = 1 tonne.
- **Capacity:** 1000 ml = 1 L; 1 cm^3 = 1 ml; 1000 cm^3 = 1 L; 1 m^3 = 1000 L.
- **Area is squared:** 1 m^2 = 10 000 cm^2 (NOT 100); 1 cm^2 = 100 mm^2.
- **Volume is cubed:** 1 m^3 = 1 000 000 cm^3.
- **Time:** 1 h = 60 min = 3600 s. To get **decimal hours** from minutes: ÷ 60.

Compound Units

- **Speed–Distance–Time:**
  `Speed = Distance ÷ Time`, `Distance = Speed × Time`, `Time = Distance ÷ Speed`.

- **Density–Mass–Volume:**
  `Density = Mass ÷ Volume`, `Mass = Density × Volume`, `Volume = Mass ÷ Density`.
- **Pressure–Force–Area:**
  `Pressure = Force ÷ Area`, `Force = Pressure × Area`, `Area = Force ÷ Pressure`.
- **Conversions to remember:**
  - km/h ↔ m/s: **÷ 3.6** (km/h → m/s), **× 3.6** (m/s → km/h).
  - 1 g/cm^3 = 1000 kg/m^3.
  - 1 N/m^2 = 1 Pa.
  - Use `g = 10 N/kg` for weight = mass × g unless told otherwise.

Percentages

- **% of an amount:** divide the % by 100, then multiply. e.g. 18% of 250 = `0.18 × 250`.
- **Increase by p%:** × `(1 + p/100)`. **Decrease by p%:** × `(1 − p/100)`.
- **Reverse %:** if you know the **final** amount, **divide** by the multiplier to get the original.
- **Compound interest:** `Final = Original × (1 + r/100)ⁿ`.
- **Percentage change:** `(new − old) ÷ old × 100`.

🎲 PROBABILITY

- **Theoretical probability** = favourable outcomes ÷ total outcomes.
- **Experimental probability** (relative frequency) = times event happened ÷ number of trials.
- **Expected frequency** = probability × number of trials.
- All probabilities sum to 1: `P(not A) = 1 − P(A)`.
- **Tree diagrams: multiply along** the branches, **add the ends** for the outcomes you want.
- Probabilities are between 0 and 1 — give as a fraction (simplest form), decimal, or percentage; **be consistent** in one question.

🧠 EXAM TIPS

- Read every question **twice** before starting; underline what is asked for.
- Show **every line of working** — most marks are for method, not the final number.
- Watch units: convert before substituting.
- Round only at the **end** (use full accuracy in working).
- After answering, sanity-check: is the answer the right size and the right unit?

✨ Good luck Shrishti — you've got this!